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Life expectancy after kidney transplantation in a population-based retrospective cohort

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Kidney transplantation is the treatment of choice for end-stage renal disease, offering significant survival benefits. However, outcomes vary based on donor type and patient characteristics. We aimed to identify factors influencing long-term survival following kidney transplantation. This was a population-based retrospective cohort study of kidney transplant recipients from 2005 to 2018 in Israel, assessing survival from one-month post-transplant. Univariate and multivariate Cox regression analyses were performed to identify predictors of mortality. Among 1,847 recipients, 679 received kidneys from living donors and 1,168 from deceased donors. The median age was 52 years (IQR 38–62), and the mean post-transplant survival time was 11.9 years (95% CI 11.7–12.3). Increased mortality risk was associated with age over 65 years (HR 3.73), current or past smoking (HR= 1.32), diabetes mellitus (HR= 1.79), heart failure (HR= 1.49), and atrial fibrillation (HR= 1.79). Receiving a kidney from a living donor (HR= 0.49) and higher pre-transplant hemoglobin levels (HR= 0.75) were linked to improved survival. Recipient comorbidities and donor type are key determinants of post-transplant outcomes. These findings support a tailored approach to transplant evaluation, balancing potential survival and quality-of-life benefits with individual risk profiles.

Keywords Kidney transplantation, Life-expectancy, Risk factors, High-risk patients, Donor type

Kidney transplantation is the treatment of choice for suitable patients with end-stage renal disease (ESRD), but it is among the most expensive medical procedures[1, 2]. In the past two decades, the number of kidney transplants has increased dramatically[3]. Advances in surgical technology and novel immunosuppressive therapies have significantly improved life expectancy following kidney transplantation, leading to a remarkable 95% one-year post-transplant survival rate[4, 5]. However, not all patients achieve the same survival and quality of life benefits from a kidney transplant, a procedure that places a high burden on the donor (living), recipients, and the healthcare system[6, 7, 8, 9, 10, 11]. The high initial healthcare costs, owing to expenses related to the pre- and post-surgical procedures (such as complications, length of hospital stay, induction agents), and high dosing of maintenance immunosuppression, are balanced out as we move away from the initial phase[12].

Life expectancy serves as a consistent and fundamental criterion for identifying individuals who may be suitable candidates for transplantation[13]. In a recently published systematic review and meta-analysis that sought to compare the long-term survival benefit between kidney transplantation and waitlist patients remaining on dialysis, transplant patients had a significant survival benefit [Hazard Ratio (HR) 0.45, 95% confidence intervals (CI) 0.39-0.54], with significant heterogeneity across studies (I^2=95.3%). Subgroup and sensitivity analyses, as well as meta-regression, did not clarify the cause of this substantial variation and failed to demonstrate the extent of survival benefit across different patient subgroups, with conflicting evidence emerging from analyses based on selected demographics[14]. Social and economic factors, such as race and socioeconomic status, also play a role in influencing survival after transplantation, with some minority groups facing higher graft failure rates[15]. Transplants from living donors generally result in better long-term outcomes compared to deceased donors, likely due to factors such as patient selection, better organ quality, shorter cold ischemia times and better surgical planning[16]. A longer duration on dialysis prior to transplantation is associated with poorer post-transplant outcomes[17].

We aimed to identify risk factors for all-cause mortality following kidney transplantation to better characterize high-risk patients and estimate their long-term life expectancy.

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Unlike previously published studies that used data available prior to transplantation, we assessed patients one-month post-transplant surgery to eliminate the influence of surgical complications and acute rejection.

Methods

Data collection and patient inclusion

This retrospective cohort study was based on data from insured members of Clalit Health Services (CHS) from the Dan Petah-Tikva and Tel Aviv districts or insured members who visited Beilinson Hospital or Hasharon Hospital. CHS is the largest healthcare delivery system in Israel, covering 53% of the population. Data were extracted anonymously from CHS using Clalit's data-sharing platform powered by MDClone (https://www.mdclone.com). Our report follows the STROBE (Strengthening the Reporting of Observational Studies in Epidemiology) guidelines[18] (Supplementary file).

All insured members aged ≥18 who underwent a single kidney transplant between 2005 and 2018 were considered for inclusion. To focus on long-term outcomes, we used a 30-day landmark and included only recipients who survived ≥30 days post-transplant. Follow-up began at the landmark (delayed entry) and continued until death or the last available electronic medical records date, including administrative censoring at 1 July 2023. Censoring therefore reflects any termination of follow-up within the Clalit electronic medical record system (loss of follow-up or insurance transition), not only administrative study end.

Of 1,890 identified recipients, 1,847 (97.7%) reached the 30-day landmark and were included; 43 (2.3%) did not (death or early graft loss). Data collection included patient demographics such as age, gender, ethnicity, peripheral index (a metric developed by the Central Bureau of Statistics, which reflects the degree of peripherality of a local authority in Israel in terms of its geographical location relative to economic activity centers) and socioeconomic status (according to the Central Bureau of Statistics is a composite score based on 14 variables that measure the socioeconomic level in four areas- demographics, education, standard of living, and employment), underlying diagnoses, Charlson Comorbidity Index[19], medications purchases, duration of kidney replacement therapy, lab results, smoking status prior to kidney transplant and type of donor (living/deceased). Additionally, details of the transplantation procedure and post-transplant follow-up information, such as cardiovascular and peripheral vascular events, medication use, and mortality, were recorded.

Statistical analysis

All statistical analyses were performed using IBM SPSS software v.28.0. Time-to-event analyses used a pre-specified 30-day landmark: the time origin (t=0) was day 30 post-transplant, and only recipients who survived ≥30 days were included. For multivariable models, we implemented delayed entry with entry at day 30. Univariate analysis was conducted using the Kaplan-Meier method to estimate the survival function over time and compare survival curves between different groups and variables. The log-rank test was used to compare survival distributions between groups and assess statistical significance. The multiple imputation technique was used to impute missing values. Statistically significant ($p < 0.05$) and clinically relevant variables were entered into a multivariable Cox proportional hazards regression model to identify independent predictors of survival after kidney transplantation. To assess model fit, the Akaike Information Criterion (AIC) was used to select the best model among five options. Lower AIC values indicate a better balance between model fit and complexity; therefore, the model with the lowest AIC was chosen for predicting mortality (supplementary table 1). Multicollinearity was assessed using a correlation matrix and variance inflation factor (VIF) analysis to evaluate the interrelationships among predictor variables. Results were expressed as hazard ratios (HR) with 95% confidence intervals (CIs).

Ethical considerations

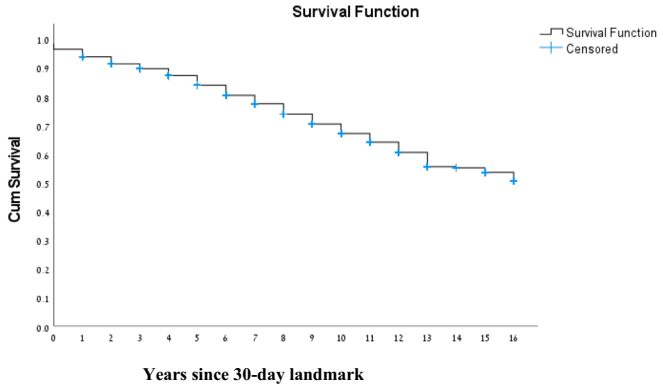
The study protocol and the waiver of informed consent were approved by the Rabin Medical Center Helsinki Committee (Institutional Review Board) (approval No. 0622-24-RMC). The study used de-identified data extracted from electronic medical records, with no direct patient contact, and was conducted in accordance with the Declaration of Helsinki and relevant regulations.

Results

Our study comprised of 1,847 kidney transplant patients; 679 received a kidney from a living donor, and 1,168 from a deceased donor. The mean post-transplant survival time was 11.9 years (95% CI 11.7–12.3) (Figure 1). Median survival time could not be estimated, as fewer than 50% of patients died during follow-up, and the survival curve did not fall below 50%.

A quarter (463/1,847) of the cohort died during the study follow-up period. Recipients' baseline characteristics are shown in Table 1 according to vital status. The median age at the time of transplant was 52 years (IQR 38-62), with 18.5% (341/1,847) of cohort over the age of 65. One third of the study cohort were female (620/1,847). The majority of kidney transplant patients were Jewish (79%, 1,400/1,765), from a medium socioeconomic class (65.7%, 1,163/1,847), and had never smoked (61.8%, 970/1,569). Most kidney transplant were on renal replacement therapy prior to kidney transplantation [83.4% (1,541/1,847)] for a median duration of 3.6 years (IQR 1.4-6.1). Chronic hypertension was the most prevalent comorbid condition prior to kidney transplantation (86%, 1,588/1,847), followed by diabetes mellitus and atherosclerotic heart disease [567/1,847 (30.7%) and 534/1,847 (28.9%), respectively] (Table 1).

In Kaplan-Meier analysis, recipients under the age of 50 had a mean life expectancy of 14 years (95%CI 13.7-14.4), those aged 50–64 had a mean life expectancy of 10.8 years (95%CI 10.4-11.4), and recipients aged 65 or older had a mean life expectancy of 7.5 years (95%CI 6.8-8.3). Chronic medical conditions prior to kidney transplantation, such as atherosclerotic heart disease, cerebral vascular accident, heart failure, peripheral



Number of patients at risk at each year of follow-up:

Years	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Patients at risk	1,847	1,779	1,718	1,496	1,289	1,111	923	775	671	543	431	337	260	197	148	105	51

* Median survival could not be estimated, as >50% of patients remained alive at the end of follow-up. Mean survival time was 11.9 years (95% CI 11.7–12.3).

** Censoring represents any termination of follow-up within the Clalit electronic medical records (including loss of follow-up or insurance transition), not only administrative censoring at 1 July 2023.

Fig. 1. Kaplan–Meier curve of overall survival from the 30-day landmark after kidney transplantation. * Median survival could not be estimated, as >50% of patients remained alive at the end of follow-up. Mean survival time was 11.9 years (95% CI 11.7–12.3). ** Censoring represents any termination of follow-up within the Clalit electronic medical records (including loss of follow-up or insurance transition), not only administrative censoring at 1 July 2023.

vascular disease, diabetes mellitus, hypertension, and atrial fibrillation, were negatively associated with long-term survival. Survival rates were higher for patients who received a kidney from a living donor. Other factors predictive of long-term survival are presented in Supplementary table 2.

Independent predictors of mortality in multivariate Cox analysis included age over 65 years [Hazard ratio (HR) 3.73, 95% confidence interval (CI) 2.82–4.92], current or past smoking (HR 1.32, 95% CI 1.07–1.64), and prior transplant comorbid conditions including diabetes mellitus (HR 1.79, 95% CI 1.47–2.19), heart failure (HR 1.49, 95% CI 1.2–1.87), and atrial fibrillation (HR 1.79, 95% CI 1.38–2.33). Receiving a kidney from a living donor (HR 0.49, 95% CI 0.38–0.62) and having high hemoglobin levels (HR 0.75, 95% CI 0.58–0.97) prior to transplantation were independent predictors of long-term survival (Table 2). Survival curves stratified by the most significant risk factors (age group, presence of diabetes, atherosclerotic heart disease, and kidney from deceased/living donor) are presented in Figure 2.

The mean life expectancy for post-transplant patients with a combination of the three strongest risk factors - age >65, diabetes mellitus, and a deceased donor was approximately four years, with less than 20% surviving beyond eight years (Supplementary figure 1).

Discussion

In this cohort of 1,847 kidney transplant patients, the mean post-transplant life expectancy was 11.9 years, with significant variations driven by donor type and patient characteristics. Key factors influencing post-transplant survival included older age, current/past smoking, and pre-existing comorbidities such as diabetes mellitus, heart failure, and atrial fibrillation were associated with increased mortality. In contrast, receiving a kidney from a living donor and having higher hemoglobin levels prior to transplantation were independent predictors of long-term survival.

Among patients with the highest risk factors such as being over the age of 65, having diabetes mellitus, and receiving a kidney from a deceased donor the median life expectancy post-transplant was 4 years, with nearly 20% surviving beyond eight years. These findings are consistent with existing literature, which shows that kidney transplantation generally confers a significant survival advantage over remaining on dialysis, despite the presence of high-risk factors for mortality[1, 20, 21, 22].

A study by De Lima et al. compared transplant recipients with dialysis patients and found that the survival advantage of transplantation was primarily observed in high-risk patients, as defined by age and comorbidities. Conversely, among younger and healthier individuals, transplantation did not provide a significant increase in life expectancy or reduce the risk of cardiovascular mortality compared to dialysis. These findings suggest that

	All cohort N=1,847	Alive by end of follow-up N=1,384	Dead by end of follow-up N=463	P-value
Patient characteristics				
Age (year)	52 (38-62)	48 (35-60)	60 (52-66)	<0.001
Gender (Female)	620 (33.6)	502 (36.3)	118 (25.5)	<0.001
BMI	26 (22.8-29.6)	25.7 (22.6-29.4)	26.8 (23.8-30.8)	<0.001
Population sector-Jewish	1400/1765 (79.3)	1033/1324 (78)	367/441 (83.2)	0.002
Peripheral area				0.254
Peripheral	301 (16.8)	224 (16.2)	77 (16.6)	
Medium	587 (31.8)	454 (32.8)	133 (28.7)	
Central	957 (51.9)	706 (51)	253 (53.7)	
Smoking status	N=1569	N=1214	N=355	0.001
never	970 (61.8)	780 (64.3)	190 (53.5)	
current	269 (17.1)	197 (16.2)	72 (20.3)	
Former	330 (21)	237 (19.5)	93 (26.2)	
Socioeconomic status	N=1771	N=1,326	N=445	0.005
Low	319 (18)	242 (18.3)	77 (17.3)	
Medium	1163 (65.7)	847 (63.9)	316 (71)	
High	289 (16.3)	237 (17.9)	52 (11.7)	
Dialysis before transplant	1541 (83.4)	1138 (82.2)	403 (87)	0.016
Dialysis vintage (years), N=1541	3.6 (1.4-6.1)	3.1 (1.2-6)	4.5 (2.9-6.4)	<0.001
Donor (Living)	679 (36.8)	591 (42.7)	88 (19)	<0.001
Transplant era				<0.001
2005-2007	261 (14.1)	143 (10.3)	118 (25.5)	
2008-2010	261 (14.1)	146 (10.5)	115 (24.8)	
2011-2013	369 (20)	257 (18.6)	112 (24.2)	
2014-2016	403 (21.8)	335 (24.2)	68 (14.7)	
2017-2019	553 (29.9)	503 (36.3)	50 (10.8)	
Comorbidities (before transplantation)				
Atherosclerotic heart disease	534 (28.9)	313 (22.6)	221 (47.7)	<0.001
Heart failure	311 (16.8)	176 (12.7)	135 (29.2)	<0.001
Cerebral vascular accident	231 (12.5)	143 (10.3)	88 (19)	<0.001
Peripheral vascular disease	236 (12.8)	103 (7.4)	133 (28.7)	<0.001
Diabetes mellitus	567 (30.7)	340 (24.6)	227 (49)	<0.001
Hypertension	1588 (86)	1162 (84)	426 (92)	<0.001
Atrial fibrillation	161 (8.7)	87 (6.3)	74 (16)	<0.001

Table 1. Baseline characteristics of the transplant recipients based on the vital status. *Patient characteristics where measured one-month post-transplant. **Categorical variables are presented as number (percentage) and continuous variables as median (interquartile range).

Risk factor	Multivariate cox regression analysis, Hazard Ratio (95% CI)	p-value
Age<50	reference	<0.001
50-64	2.13 (1.66-2.73)	
>=65	3.73 (2.82-4.92)	
Donor (living)	0.49 (0.38-0.62)	<0.001
Diabetes mellitus	1.79 (1.47-2.19)	<0.001
Heart failure	1.49 (1.20-1.87)	<0.001
Cerebral vascular accident	1.16 (0.91-1.48)	0.238
Atherosclerotic heart disease	1.24 (1.00-1.53)	0.050
Atrial fibrillation	1.79 (1.38-2.33)	<0.001
Hypertension	1.17 (0.83-1.67)	0.373
Smoker	1.32 (1.07-1.64)	0.009
Hemoglobin >11 g/dL	0.75 (0.58-0.97)	0.027

Table 2. Independent predictors of long-term survival after kidney transplantation: multivariable Cox proportional hazards model (N = 1,847).

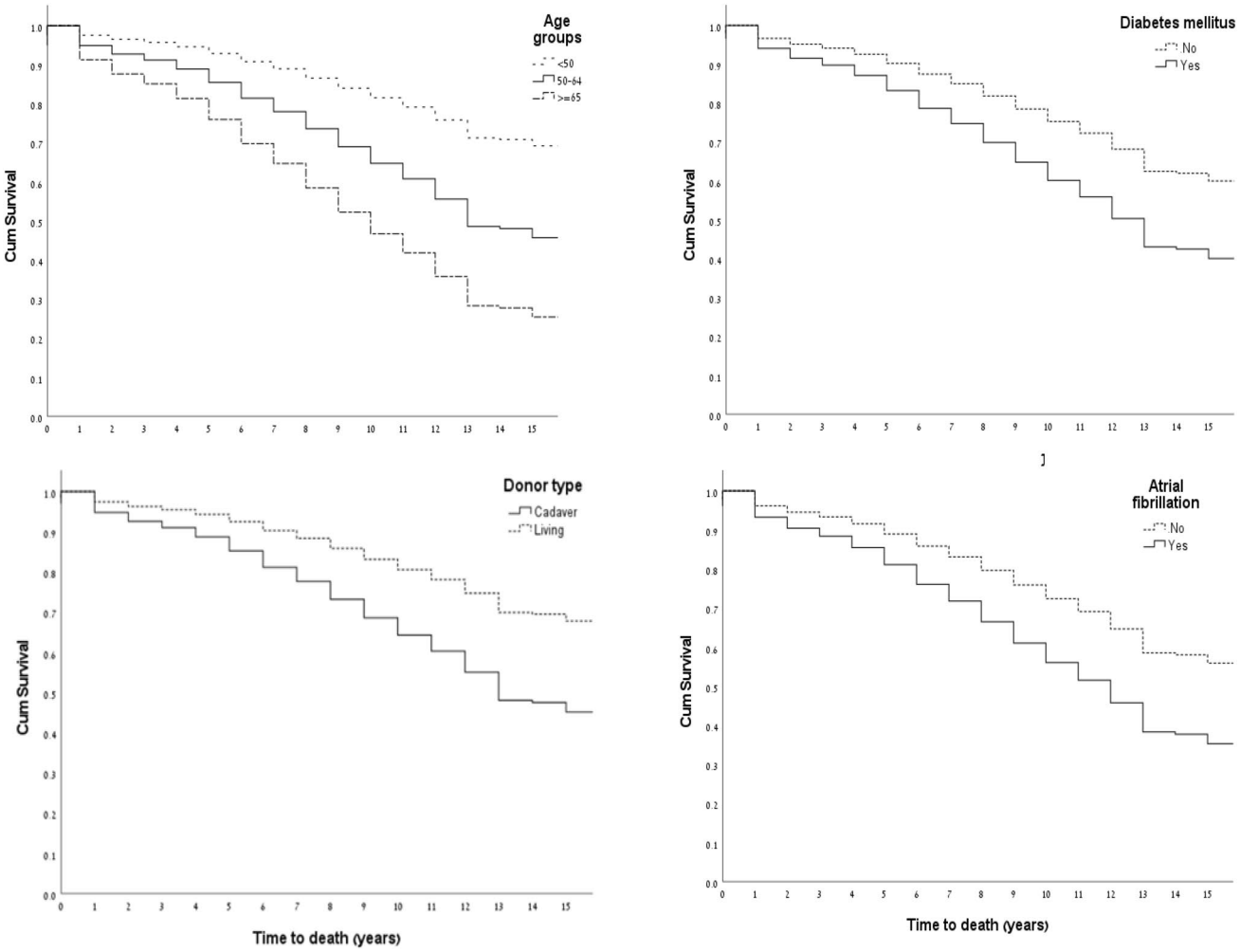


Fig. 2. Cox regression survival curves stratified by age >70, donor type (living vs. deceased), presence of diabetes mellitus, and atrial fibrillation.

age and comorbidities should not be exclusionary factors; rather, they should serve as strong indications for early transplantation[10]. High-risk individuals may have shorter overall survival compared to low-risk patients, but their relative benefit from transplantation is greater, offering meaningful improvements in both life expectancy and quality of life.

These findings align with prior literature suggesting that high-risk patient, those over 65 or with significant comorbidities, may derive meaningful relative benefit from transplantation compared to remaining on dialysis[10]. Although our study did not directly evaluate dialysis outcomes, our findings support a nuanced approach to transplant eligibility criteria, recognizing that patients with shorter projected survival may still experience clinically relevant gains in both life expectancy and quality of life. Our study reinforces the critical role of living donors in improving outcomes. Transplants from living donors were associated with significantly better survival compared to those from deceased donors. These findings contribute to the broader discussion on optimizing donor selection criteria and emphasize the significant benefits of prioritizing living donor transplants when possible[23]. Moreover, preemptive transplantation from living donors offers a unique advantage by avoiding the cumulative complications of prolonged dialysis, such as vascular damage and cardiovascular decline, further enhancing long-term survival and quality of life[24, 25].

Our analysis focused on patients who survived beyond 30-days post-transplant, enabling us to examine predictors of longer-term survival in a clinically stable cohort. While this approach excludes early mortality events, it allows for clearer identification of long-term risk factors without the confounding impact of immediate post-surgical complications. This design, similar to landmark strategies used in prior studies[5], supports more focused evaluation of long-term outcomes while recognizing the inherent limitations of excluding early mortality events which warrant separate investigation.

While our study offers valuable insights, several limitations must be acknowledged. Detailed data on donor characteristics and comorbidities were not available in the Clalit dataset, which could influence transplant outcomes. The increasing use of less-than-optimal quality donors, such as older donors or those with comorbid conditions, might have an impact on the analysis, particularly in analyses involving deceased donors in

Eurotransplant, for example, the median age of deceased kidney donors has risen significantly, and about one-third of donors can now be classified as expanded criteria donors[26]. Nevertheless, our findings remain consistent with prior literature, and future studies utilizing enriched registries or institutional databases may provide additional insights. Another limitation is the absence of detailed data on graft loss or return to dialysis, which were not available in the dataset used for this analysis. Our outcome assessment focused on all-cause mortality, with complete follow-up available for all patients through the Clalit Health Services electronic medical records. Findings generalize to 30-day survivors and do not address immediate postoperative outcome.

In conclusion, long-term survival after kidney transplantation is strongly influenced by both donor characteristics and recipient comorbidities. Our findings highlight the importance of individualized risk assessment in transplant decision-making, particularly for older patients and those with significant cardiovascular burden. While transplantation may offer potential benefits even for high-risk recipients, careful patient selection and management remain essential to optimizing outcomes. Consistent with prior research, we advocate for the expansion of living donor programs, including altruistic donations, and the development of tailored selection criteria to maximize transplantation benefits and address the diverse needs of patients with chronic kidney disease. Furthermore, our findings suggest an important direction for future research: exploring whether improved management of diabetes, atrial fibrillation, heart failure, and hypertension can further enhance transplant outcomes.

Data availability

The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

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Author contributions

1. Tanya Babich - Participated in research design - Participated in data analysis - Participated in the writing of the paper 2. Vered Daitch - Participated in the performance of the research - Participated in data analysis 3. Leonard Leibovici - Participated in research design - Contributed to the writing and critical revision of the paper 4. Hefziba Green^{ } - Participated in research design - Contributed to the critical revision of the paper 5. Adi Turjeman - Contributed to the performance of the research - Assisted with the writing of the paper.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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